

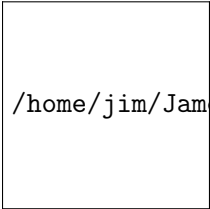
Section 2.1: Sets and Set Operations

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Wednesday 20th August, 2014 and Friday 22nd August, 2014



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Chapter 2: Sets and Counting

2.1 Sets and Set Operations

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2.2 Applications of Venn Diagrams

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2.2 Applications of Venn Diagrams

2.3 Intro to Combinatorics

Chapter 2: Sets and Counting

- 2.1 Sets and Set Operations
- 2.2 Applications of Venn Diagrams
- 2.3 Intro to Combinatorics
- 2.4 Permutations & Combinations

ex. $\{0,2,4\}$

Sets

Set

A *set* is a well defined collection of distinct objects.

ex. $\{0, 2, 4\}$

ex. $R = \{Billy, Bob, Bubba\}$

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- Usually written as capital letters, e.g., A , B , C , etc.

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The objects in the set are called *elements* or *members*.

- $x \in A$ means x is an element in A .
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ex. $3.14 \notin \{1, 2, 3, 4, \dots\}$

ex. $x \notin \emptyset$

Two sets are *equal* if they contain the same elements.

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ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

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Do not write $\{0, 2, 2, 4\}$

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ex. $\{1, 2, 3, 4, \dots\}$

ex. $\{x \mid x > 0 \text{ and } x \text{ is an integer}\}$

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The *cardinal number* of a set A is the number of elements in A .

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The *cardinal number* of a set A is the number of elements in A .

- Written, $n(A)$.

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ex. $n(\{0, 2, 4\}) = 3$

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The *cardinal number* of a set A is the number of elements in A .

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ex. $n(\{0, 2, 4\}) = 3$

ex. If $A = \{a, b\}$ then $n(A) = 2$.

Problem 2.1.1. Write the following set in *tabular* and *set-builder* form.

The set of even, positive integers.

Problem 2.1.2. Let $A = \{1, 2, 3, 4\}$, $B = \{4, 3, 1, 2\}$, and $C = \{1, 2, 3\}$

True or False: $A = B$

Problem 2.1.3. Let $A = \{1, 2, 3, 4\}$, $B = \{4, 3, 1, 2\}$, and $C = \{1, 2, 3\}$

True or False: $4 \notin A$

Problem 2.1.4. Let $A = \{1, 2, 3, 4\}$, $B = \{4, 3, 1, 2\}$, and $C = \{1, 2, 3\}$

Find $n(C)$

Problem 2.1.5. Decide whether the following are *true* or *false*.

$$\{a, b, \dots, z\} = \{b, c, \dots, z\}$$

Problem 2.1.6. Let $\mathbb{Z}^+ = \{1, 2, \dots\}$ and $\mathbb{N} = \{0, 1, 2, \dots\}$

True or False: $\mathbb{Z}^+ \neq \mathbb{N}$

Problem 2.1.7. Let $\mathbb{Z}^+ = \{1, 2, \dots\}$ and $\mathbb{N} = \{0, 1, 2, \dots\}$

True or False: $n(\mathbb{Z}^+) = n(\mathbb{N})$

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a set A is a *subset* of a set B if every element in A is also in B .

$A \subseteq B$: A is a subset of B

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ex. $A = \{3, 4, 5, 6\}$
 $B = \{2, 3, 4, 5, 6, 7, 8\}$
 $C = \{3, 4, 5, 6, 9\}$

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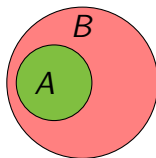
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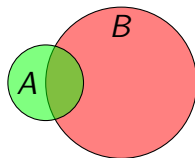
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$$A \subseteq B$$



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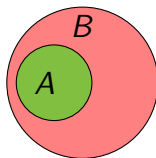
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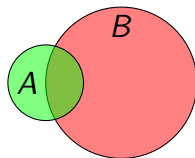
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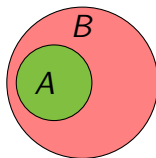
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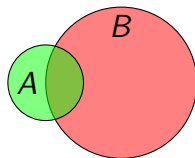
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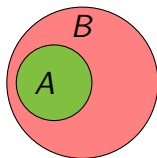
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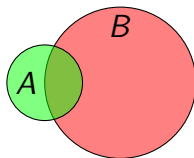
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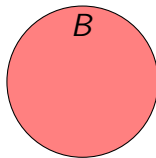
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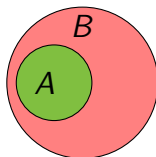
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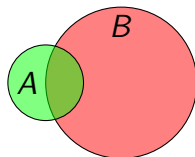
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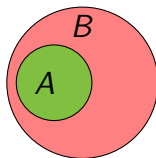
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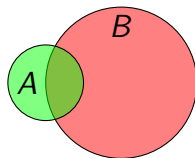
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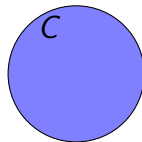
- $A \subseteq B$
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- $A \subseteq A$, $B \subseteq B$, $C \subseteq C$



$$A \subseteq B$$



$$A \not\subseteq B$$



$$C \subseteq C$$

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a set A is a *proper subset* of a set B if every element in A is also in B , and $A \neq B$.

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$A \subset B$: A is a proper subset of B .

$A \not\subset B$: $A \not\subseteq B$ and so A is not a proper subset of B

ex.

$$A = \{3, 4, 5, 6\}$$

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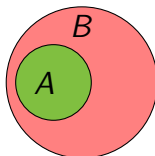
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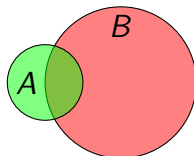
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- $A \subset B$ and $C \not\subset B$



$$A \subset B$$



$$A \not\subset B$$

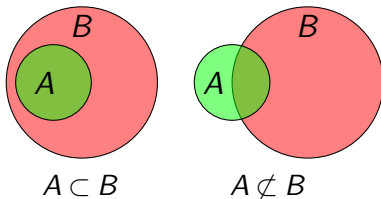
Subsets

Proper Subset

a set A is a *proper subset* of a set B if every element in A is also in B , and $A \neq B$.

$A \subset B$: A is a proper subset of B .

$A \not\subset B$: $A \not\subseteq B$ and so A is not a proper subset of B



ex.

$$A = \{3, 4, 5, 6\}$$

$$B = \{2, 3, 4, 5, 6, 7, 8\}$$

$$C = \{3, 4, 5, 6, 9\}$$

- $A \subset B$ and $C \not\subset B$
- B is not a proper subset B

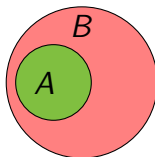
Subsets

Proper Subset

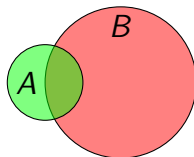
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$A \subset B$



$A \not\subset B$

ex.

$$A = \{3, 4, 5, 6\}$$

$$B = \{2, 3, 4, 5, 6, 7, 8\}$$

$$C = \{3, 4, 5, 6, 9\}$$

- $A \subset B$ and $C \not\subset B$
- B is not a proper subset B

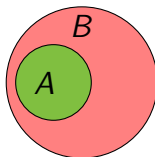
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Proper Subset

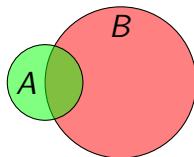
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$A \subset B$



$A \not\subset B$

ex.

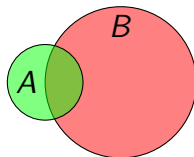
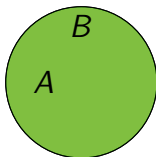
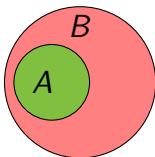
$$A = \{3, 4, 5, 6\}$$

$$B = \{2, 3, 4, 5, 6, 7, 8\}$$

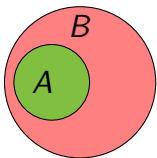
$$C = \{3, 4, 5, 6, 9\}$$

- $A \subset B$ and $C \not\subset B$
- B is not a proper subset B

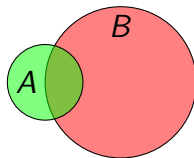
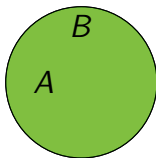
Venn Diagrams



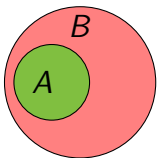
Venn Diagrams



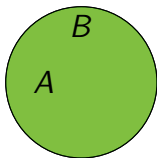
$A \subseteq B$ and $A \subset B$



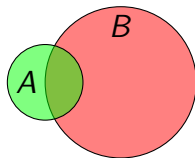
Venn Diagrams



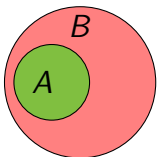
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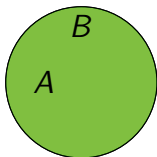
$A \subseteq B$



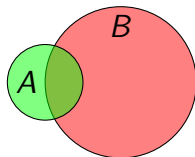
Venn Diagrams



$$A \subseteq B \text{ and } A \subset B$$

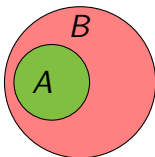


$$A \subseteq B$$

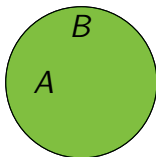


$$A \not\subseteq B$$

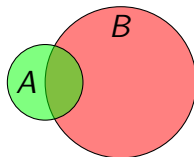
Venn Diagrams



$A \subseteq B$ and $A \subset B$



$A \subseteq B$

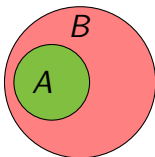


$A \not\subseteq B$

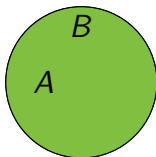
Problem 2.1.8. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$, $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$A ___ E$

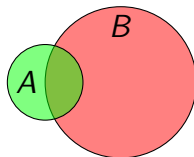
Venn Diagrams



$A \subseteq B$ and $A \subset B$



$A \subseteq B$

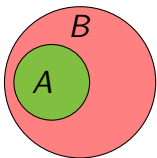


$A \not\subseteq B$

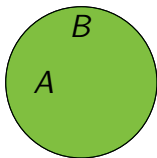
Problem 2.1.9. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$,
 $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$.
Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$\emptyset \underline{\hspace{0.5cm}} A$

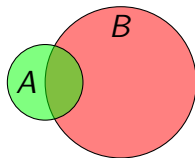
Venn Diagrams



$A \subseteq B$ and $A \subset B$



$A \subseteq B$

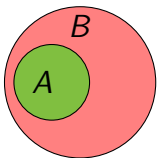


$A \not\subseteq B$

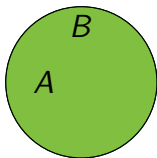
Problem 2.1.10. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$, $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$A ___ U$

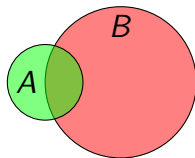
Venn Diagrams



$A \subseteq B$ and $A \subset B$



$A \subseteq B$



$A \not\subseteq B$

Problem 2.1.11. Let $A = \{2, 4, 6, 10\}$, $B = \{2, 4, 8, 10\}$, $C = \{4, 8, 12\}$, $D = \{2, 10\}$, $E = \{6\}$, and $U = \{2, 4, 6, 8, 10, 12, 14\}$. Insert $\subseteq$ or $\not\subseteq$ to make the statement true.

$A ___ U$

Set Operations

Set Operations

Set Operations

Union

The *union* of A and B is the set of all elements in A or B

Set Operations

Union

The *union* of A and B is the set of all elements in A or B

- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection

The *intersection* of A and B is the set of all elements in A and B

Set Operations

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The *union* of A and B is the set of all elements in A or B

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The *intersection* of A and B is the set of all elements in A and B

- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Set Operations

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Set Operations

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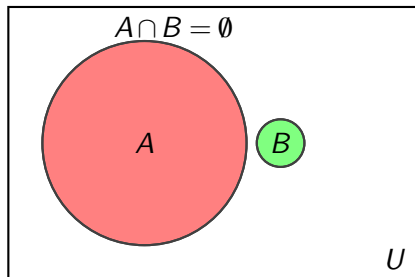
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Set Operations

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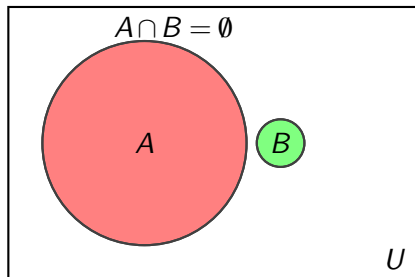
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A and B are *mutually disjoint* if $A \cap B = \emptyset$.

Set Operations

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The *union* of A and B is the set of all elements in A or B

- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

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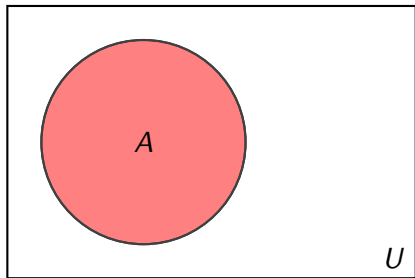
The *intersection* of A and B is the set of all elements in A and B

- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

complement

The *complement* of A is the set of everything in U but not in A .

- $A' = \{x \mid x \notin A \text{ and } x \in U\}$



A and B are *mutually disjoint* if $A \cap B = \emptyset$.

Set Operations

Union

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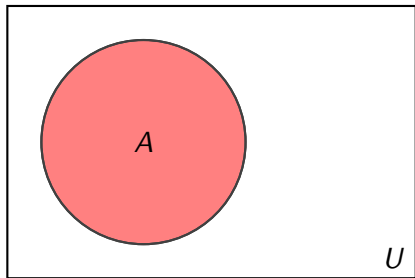
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The *complement* of A is the set of everything in U but not in A .

- $A' = \{x \mid x \notin A \text{ and } x \in U\}$



The *universal set*, denoted U , is the set of all possible elements of any set used in the problem.

Set Operations

Union

The *union* of A and B is the set of all elements in A or B

- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection

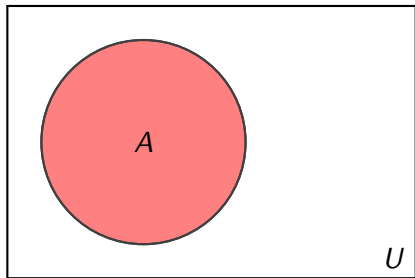
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Set Operations

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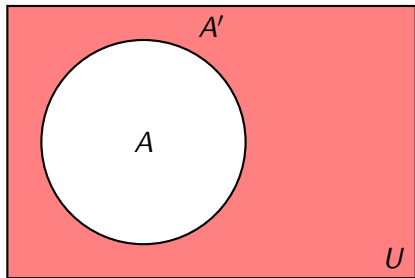
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- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

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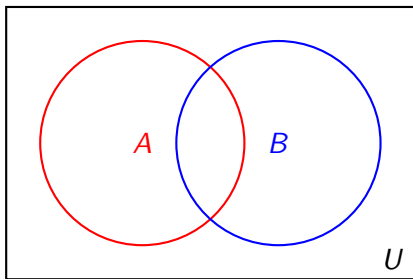
- $A' = \{x \mid x \notin A \text{ and } x \in U\}$



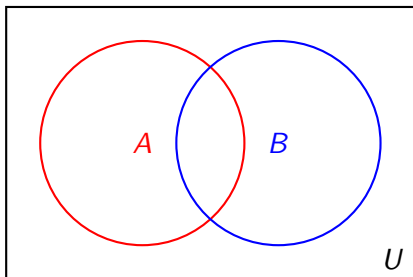
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Set Operations

Set Operations



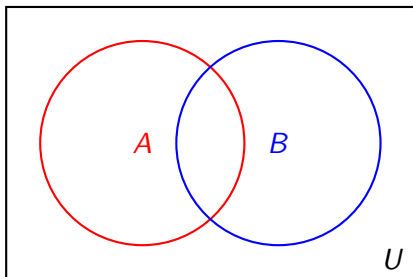
Set Operations



Problem 2.1.14. Let U be the days of the week, $A = \{M, Tu, W, Th, F\}$ and $B = \{F, Sa, M\}$. Find the indicated set.

$$(A \cap B)'$$

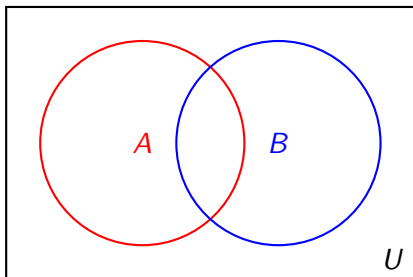
Set Operations



Problem 2.1.15. Let U be the days of the week, $A = \{M, Tu, W, Th, F\}$ and $B = \{F, Sa, M\}$. Find the indicated set.

$$A' \cup B$$

Set Operations



Problem 2.1.16. Let U be the days of the week, $A = \{M, Tu, W, Th, F\}$ and $B = \{F, Sa, M\}$. Find the indicated set.

$$A \cup B'$$

Cardinal Number Formulas

Cardinal Number Formulas

- What is U ?
- How many do not have a DVD player?

Cardinal Number Formulas

Example 2.1.1.

A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

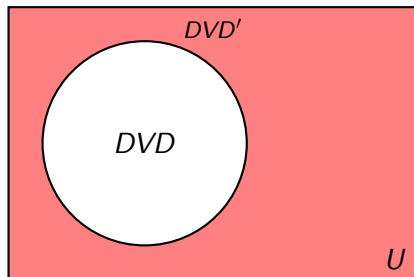
- What is U ?
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Cardinal Number Formulas

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A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

- What is U ?
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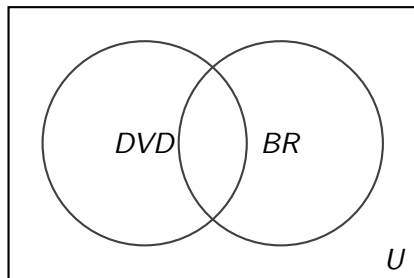
Union Rule

Cardinal Number Formulas

Example 2.1.1.

A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

- What is U ?
- How many do not have a DVD player?
- How many have neither a DVD or Blue Ray player?



Union Rule

- $$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

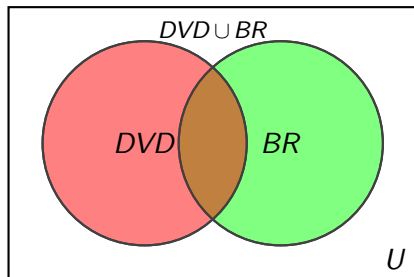
Recall: $n(A)$ is the number of elements in A .

Cardinal Number Formulas

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A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

- What is U ?
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Union Rule

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cap B) = n(A \cup B) - n(A) - n(B)$

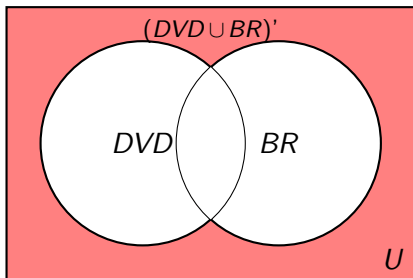
Recall: $n(A)$ is the number of elements in A .

Cardinal Number Formulas

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A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

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Union Rule

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cap B) = n(A \cup B) - n(A) - n(B)$

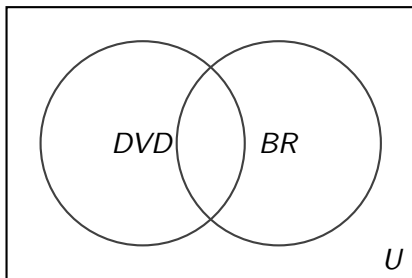
Recall: $n(A)$ is the number of elements in A .

Cardinal Number Formulas

Example 2.1.1.

A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

- What is U ?
- How many do not have a DVD player?
- How many have neither a DVD or Blue Ray player?
- How many have a Blue Ray but not a DVD player?



Union Rule

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cap B) = n(A \cup B) - n(A) - n(B)$

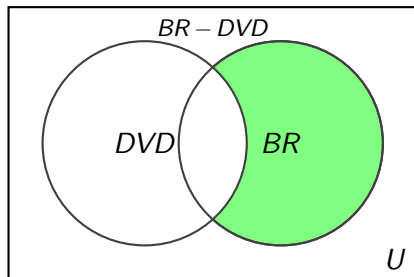
Recall: $n(A)$ is the number of elements in A .

Cardinal Number Formulas

Example 2.1.1.

A researcher collecting data on 100 houses finds that 76 have a DVD player, 21 have a Blue Ray player, and 12 have both.

- What is U ?
- How many do not have a DVD player?
- How many have neither a DVD or Blue Ray player?
- How many have a Blue Ray but not a DVD player?

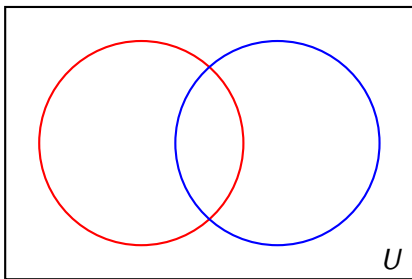


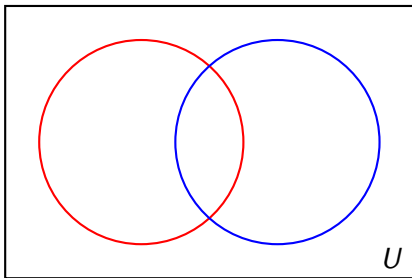
Union Rule

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cap B) = n(A \cup B) - n(A) - n(B)$

Recall: $n(A)$ is the number of elements in A .

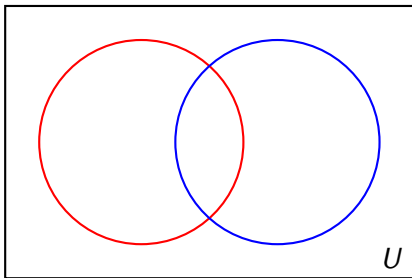
Problems





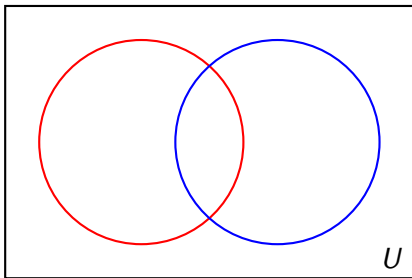
Problem 2.1.19. In a recent survey of 500 high school senior, it was found that 102 owned an automobile, 147 owned a motorcycle, and 21 owned both.

Draw a Venn diagram illustrating the results of the survey.



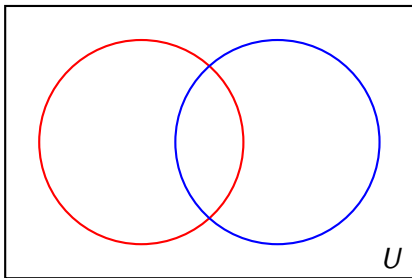
Problem 2.1.20. In a recent survey of 500 high school senior, it was found that 102 owned an automobile, 147 owned a motorcycle, and 21 owned both.

What percent of these students own an automobile or a motorcycle.



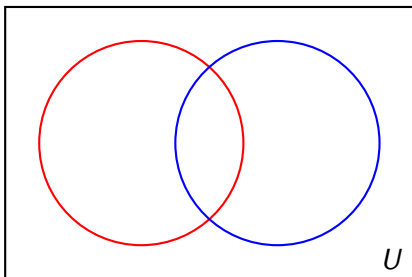
Problem 2.1.21. In a recent survey of 500 high school senior, it was found that 102 owned an automobile, 147 owned a motorcycle, and 21 owned both.

What percent of these students own an automobile and a motorcycle.



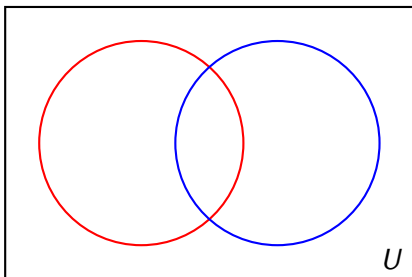
Problem 2.1.22. Determine how many cards, in an ordinary deck of playing cards, fit the description.

Spades or aces.



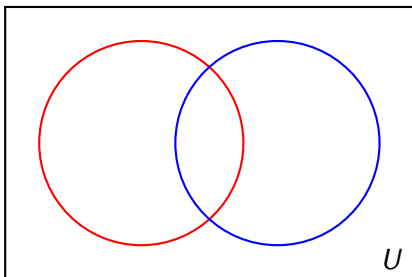
Problem 2.1.23. Determine how many cards, in an ordinary deck of playing cards, fit the description.

Face cards and red.



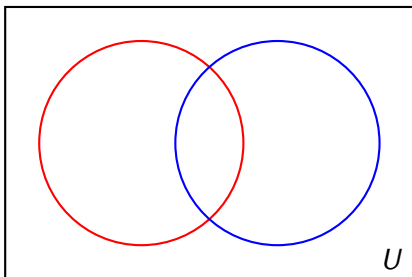
Problem 2.1.24. Determine how many cards, in an ordinary deck of playing cards, fit the description.

Face cards and red.



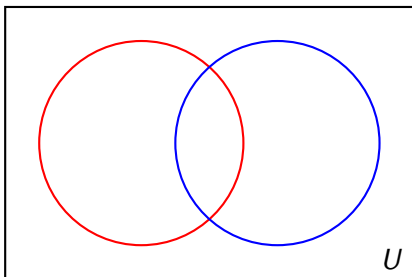
Problem 2.1.25. Answer the following.

If $A \cap B = \emptyset$, what is the relationship between A and B.



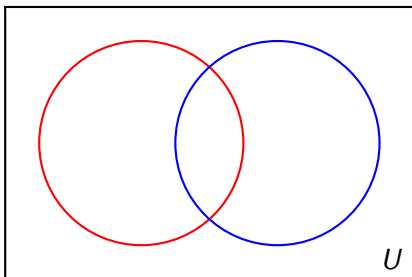
Problem 2.1.26. Answer the following.

If $A \cup B = \emptyset$, what is the relationship between A and B.



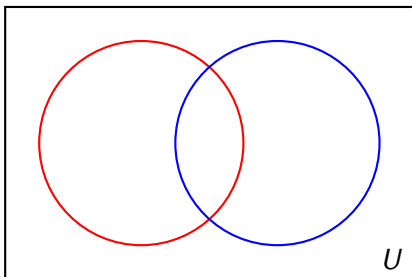
Problem 2.1.27. Answer the following.

If $A \cap B = B$, what must be true concerning these two sets.



Problem 2.1.28. Answer the following.

If $A \cup B = B$, what must be true concerning these two sets.



Problem 2.1.29. Answer the following.

If $A \cap A = \emptyset$, what must be true concerning A .